



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

26-03-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	D	D	C	D	B	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	C	D	D	A	D	A	B	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	3	5	30	25	3	640	4	30	16	6

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	D	C	A	A	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	B	D	C	C	A	B	D	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	57	6	27	5	85	49	1	5	354	77

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	D	A	B	D	A	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	A	B	B	B	B	A	C	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	2	5	41	144	68	4	2020	486	1	0

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (D)**

$$v_0 = \sqrt{2gh}$$

$$v = e\sqrt{2gh} = \sqrt{2gh}$$

$$\Rightarrow e = 0.9$$

$$S = h + 2e^2h + 2e^4h + \dots$$

$$t = \sqrt{\frac{2h}{g}} + 2e\sqrt{\frac{2h}{g}} + 2e^2\sqrt{\frac{2h}{g}} + \dots$$

$$v_{av} = \frac{S}{t} = 2.5 \text{ m/s}$$

2. **Ans (B)**

$$f = 4 + 3 + 3 = 10$$

assuming non linear

$$\beta = \frac{C_p}{C_v} = 1 + \frac{2}{f} = \frac{12}{10} = 1.2$$

3. **Ans (C)**

$$A = A_0 e^{-\gamma t}$$

$$\ln 2 = \frac{b}{2m} \times 120$$

$$\frac{0.693 \times 2 \times 1}{120} = b$$

$$1.16 \times 10^{-2} \text{ kg/sec.}$$

5. **Ans (D)**

$$P = h\rho g$$

$$\beta = \frac{p}{\frac{\Delta v}{v}} = \frac{2 \times 10^3 \times 10^3 \times 9.8}{1.36 \times 10^{-2}}$$

$$= 1.44 \times 10^9 \text{ N/m}^2$$

6. **Ans (C)**

$$T \propto R^{3/2}$$

$$\frac{24}{T} = \left(\frac{12R}{3R}\right)^{3/2} \Rightarrow T = 3 \text{ hr}$$

7. **Ans (D)**

$$\omega = 2\pi f$$

$$= 1.5 \times 10^3$$

$$A = \frac{6}{2} = 3 \text{ cm} = 0.03 \text{ m}$$

8. **Ans (B)**

Option (a) is wrong ; since in adiabatic process $V \neq \text{constant}$.

Option (b) is wrong, since in isothermal process $T = \text{constant}$

Option (c) & (d) matches isothermes & adiabatic formula :

$$TV^{\gamma-1} = \text{constant} \& \frac{T^\gamma}{p^{\gamma-1}} = \text{constant}$$

9. **Ans (B)**

$$v = v_0 + gt + Ft^2$$

$$\frac{ds}{dt} = v_0 + gt + Ft^2$$

$$\int ds = \int_0^1 (v_0 + gt + Ft^2) dt$$

$$s = \left[v_0 t + \frac{gt^2}{2} + \frac{Ft^3}{3} \right]_0^1$$

$$s = v_0 + \frac{g}{2} + \frac{F}{3}$$

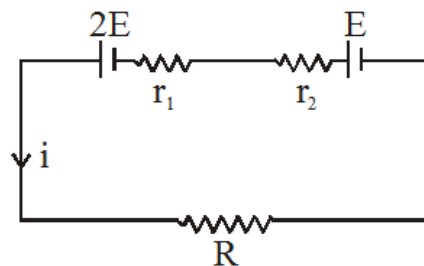
10. **Ans (D)**

$$\text{Band width} = 2 f_m$$

$$\omega_m = 1.57 \times 10^8 = 2\pi f_m$$

$$BW = 2f_m = \frac{10^8}{2} \text{ Hz} = 50 \text{ MHz}$$

11. **Ans (B)**



$$i = \frac{3E}{R + r_1 + r_2}$$

$$\text{TPD} = 2E - ir_1 = 0$$

$$2E = ir_1$$

$$2E = \frac{3E \times r_1}{R + r_1 + r_2}$$

$$2R + 2r_1 + 2r_2 = 3r_1$$

$$R = \frac{r_1}{2} - r_2$$

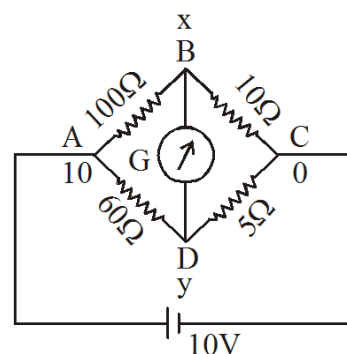
12. **Ans (B)**

$$B = 2 \times B_{\text{st.wire}} + B_{\text{loop}}$$

$$B = 2 \times \frac{\mu_0 i}{4\pi r} + \frac{\mu_0 i}{2r} \left(\frac{\pi}{2\pi} \right)$$

$$B = \frac{\mu_0 i}{4\pi r} (2 + \pi)$$

13. **Ans (C)**



$$\frac{x-10}{100} + \frac{x-y}{15} + \frac{x-0}{10} = 0$$

$$53x - 20y = 30 \dots\dots(1)$$

$$\frac{y-10}{60} + \frac{y-x}{15} + \frac{y-0}{5} = 0$$

$$17y - 4x = 10 \dots\dots(2)$$

on solving (1) & (2)

$$x = 0.865$$

$$y = 0.792$$

$$\Delta V = 0.073 R = 15\Omega$$

$$i = 4.87 \text{ mA}$$

14. **Ans (D)**

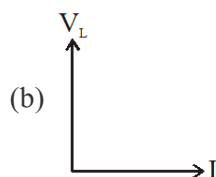
$$A_1 \omega_1 = A_2 \omega_2$$

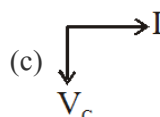
$$A_1 \sqrt{\frac{k_1}{m}} = A_2 \sqrt{\frac{k_2}{m}}$$

$$\frac{A_1}{A_2} = \sqrt{\frac{k_2}{k_1}}$$

15. **Ans (D)**

(a) 

(b) 

(c) 

(d) $\tan \phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$

16. **Ans (A)**

C comes to rest

$$V_{cm} \text{ of A \& B} = \frac{v}{2}$$

$$\Rightarrow \frac{1}{2} \text{ is } v_{ret}^2 = \frac{1}{2} kx^2$$

$$x = \sqrt{\frac{\mu \times v^2}{k}} = \sqrt{\frac{m}{2k}} v$$

18. **Ans (A)**

$$\frac{1}{2} m v_1^2 = h f_1 - \phi$$

$$\frac{1}{2} m v_2^2 = h f_2 - \phi$$

$$v_1^2 - v_2^2 = \frac{2h}{m} (f_1 - f_2)$$

19. **Ans (B)**

$$X_L = \omega L$$

$$i = \frac{V_0}{\omega L}$$

20. **Ans (C)**

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{5}{7} \times \frac{10}{2} = \frac{25}{7}$$

$$t = \frac{2v_0}{a} = \frac{2 \times 1 \times 7}{25} = 0.56$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (3)**

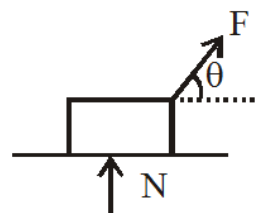
$$c \epsilon_0 E^2 = \frac{100}{4\pi \times 3^2}$$

$$c \epsilon_0 \left(\sqrt{\frac{x}{5}} E \right)^2 = \frac{60}{4\pi \times 3^2}$$

$$\Rightarrow \frac{x}{5} = \frac{3}{5}$$

$$\Rightarrow x = 3$$

2. **Ans (5)**



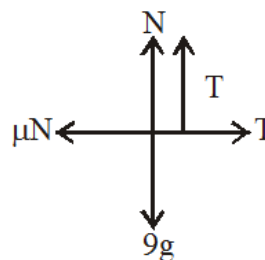
$$F \cos \theta = \mu N$$

$$F \sin \theta + N = mg$$

$$\Rightarrow F = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$$

$$F_{min} = \frac{\mu mg}{\sqrt{1 + \mu^2}} = \frac{\frac{1}{\sqrt{3}} \times 10}{\frac{2}{\sqrt{3}}} = 5$$

3. **Ans (30)**



$$N + T = 90$$

$$T = \mu N = 0.5 (90 - T)$$

$$1.5 T = 45$$

$$T = 30$$

4. **Ans (25)**

$$4t_T = 100 \times \frac{4}{3} \pi r^3$$

$$= 100 \times \frac{4\pi}{3} \times \frac{3}{40\pi} \times 10^{-9} = 10^{-8} \text{ cm}^3$$

$$t_T = 25 \times 10^{-10} \text{ cm}$$

$$= 25 \times 10^{-12} \text{ m}$$

$$t_0 = 0.01 \text{ t}_T = 25 \times 10^{-14} \text{ m}$$

$$= 25$$

PART-2 : CHEMISTRY

SECTION-I

5. **Ans (3)**

$$F = \frac{-dU}{dr} = -4U_0 r^3 = \frac{mv^2}{r}$$

$$mv^2 = 4U_0 r^4$$

$$v \propto r^2$$

$$mvr = \frac{nh}{2\pi}$$

$$r^3 \propto n$$

$$r \propto n^{1/3} = 3$$

6. **Ans (640)**

$$\phi = E_x A \Rightarrow \frac{2}{5} \times 4 \times 10^3 \times 0.4 = 640$$

7. **Ans (4)**

$$\text{C.O.M of quarter disc is at } \frac{4a}{3\pi}, \frac{4a}{3\pi} = 4$$

8. **Ans (30)**

$$\lambda_m = \frac{\lambda_a}{\mu} \Rightarrow \mu = \frac{3}{2}$$

$$\frac{\mu}{v} - \frac{1}{u} = \frac{\mu - 1}{R}$$

$$\frac{3}{2 \times 10} + \frac{1}{15} = \frac{\frac{3}{2} - 1}{R}$$

$$R = \frac{30}{13} = 30$$

9. **Ans (16)**

$$20 = (C_1 + C_2) V \Rightarrow V = 2 \text{ volt.}$$

$$Q_2 = C_2 V = 16 \mu\text{CF}$$

$$= 16$$

10. **Ans (6)**

$$J_c = \frac{E}{\rho} = \frac{V}{\rho d}$$

$$J_d = \frac{1}{A} \frac{dq}{dt}$$

$$= \frac{C}{A} \frac{dV_c}{dt} = \frac{\epsilon}{d} \frac{dV_c}{dt}$$

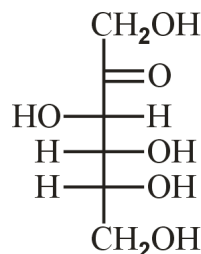
$$\Rightarrow \frac{V_0 \sin 2\pi ft}{\rho d} = 10^x \times \frac{80\epsilon_0}{d} V_0 (2\pi f) \cos 2\pi ft$$

$$\tan \left(2\pi \times \frac{900}{800} \right) = 10^x \times \frac{40}{9 \times 10^9} \times 900$$

$$= x = 6$$

1. **Ans (B)**

Fructose is a ketohexose.



2. **Ans (D)**

Li-Mg, B-Si, Be-Al show diagonal relationship but Li and Na do not show diagonal relationship as both belongs to same group and not placed diagonally.

3. **Ans (A)**

Cation exchanger contains $-\text{SO}_3\text{H}$ or $-\text{COOH}$ groups while anion exchanger contains basic groups like $-\text{NH}_2$.

4. **Ans (D)**

Ore	Formula
(a) Haematite	Fe_2O_3
(b) Bauxite	$\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
(c) Magnetite	Fe_3O_4
(d) Malachite	$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$

5. **Ans (C)**

Ambident nucleophile

(A) KCN & AgCN

(C) AgNO_2 & KNO_2

6. **Ans (A)**

N_2O and NO are neutral oxides of nitrogen NO_2 and N_2O_3 are acidic oxides.

7. Ans (A)

Complex		Type of Isomerism	
(a)	$[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$	(i)	Co-ordination isomerism
(b)	$[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$	(ii)	Linkage isomerism
(c)	$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$	(iii)	Solvate isomerism
(d)	cis- $[\text{CrCl}_2(\text{ox})_2]^{3-}$	(iv)	Optical isomerism

8. Ans (A)

Primary amines react with Para Toluene sulfonyl chloride to form a precipitate that is soluble in NaOH.

Secondary amines reacts with para toluene sulfonyl chloride to give a precipitate that is insoluble in NaOH.

Tertiary amines do not react with para toluene.

9. Ans (A)

Cr(Z=24)

$[\text{Ar}] 4s^1 3d^5$ Cr shows common oxidation states starting from +2 to +6.

10. Ans (B)

Artificial sweetener : Sucralose

Antiseptic : Bithional

Preservative : Sodium Benzoate

Glyceryl ester of stearic acid : Sodium stearate

11. Ans (C)

Alkane are very less reactive, tertiary hydrogen can oxidise to alcohol with KMnO_4 .

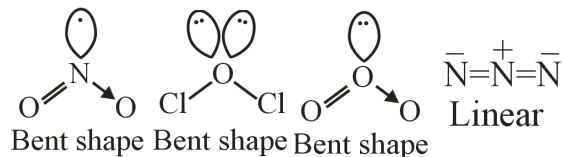


2-methyl-butane

12. Ans (B)

Kjeldahl method is not applicable to compounds containing nitrogen in nitro group, Azo groups and nitrogen present in the ring (e.g Pyridine) as nitrogen of these compounds does not change to Ammonium sulphate under these conditions.

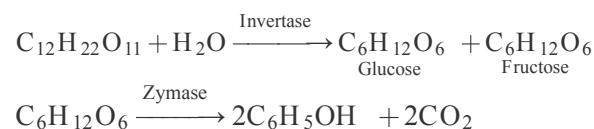
13. Ans (D)



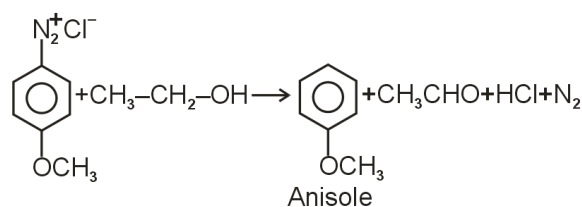
14. Ans (C)

Informative

OR



16. Ans (A)



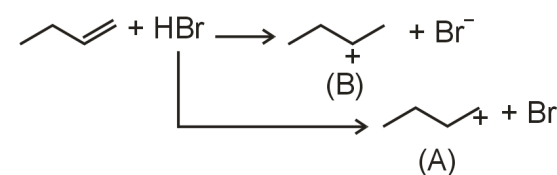
17. Ans (B)

To coagulate negative sol, cation with higher charge has higher coagulation value.

18. Ans (D)

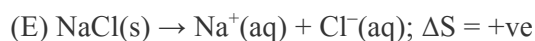
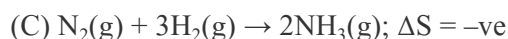
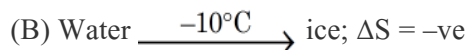
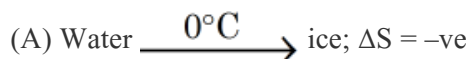
The process in which nutrient enriched water bodies support a dense plant population which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity is known as eutrophication.

19. Ans (A)



This is more stable due to secondary cation formation and formed with faster rate due to low activation energy.

20. Ans (A)



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (57)**

$$\kappa = \frac{1}{R} \cdot G^*$$

For same conductivity cell, G^* is constant and hence $\kappa \cdot R = \text{constant}$.

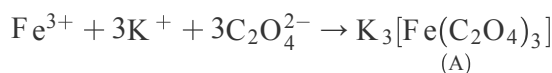
$$\therefore 0.14 \times 4.19 = \kappa \times 1.03$$

$$\text{or, } \kappa \text{ of HCl solution} = \frac{0.14 \times 4.19}{1.03}$$

$$= 0.5695 \text{ Sm}^{-1}$$

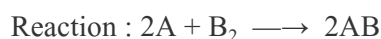
$$= 56.95 \times 10^{-2} \text{ Sm}^{-1} \approx 57 \times 10^{-2} \text{ Sm}^{-1}$$

2. **Ans (6)**



Secondary valency of Fe in 'A' is 6.

3. **Ans (27)**



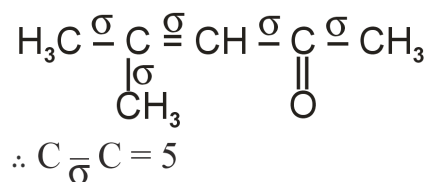
As the reaction is elementary, the rate of reaction is

$$r = K \cdot [\text{A}]^2 [\text{B}_2]$$

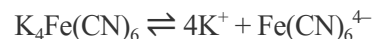
on reducing the volume by a factor of 3, the concentrations of A and B_2 will become 3 times and hence, the rate becomes $3^2 \times 3 = 27$ times of initial rate.

4. **Ans (5)**

Mesitylene oxide



5. **Ans (85)**



$$\text{Initial conc. } 1 \text{ m} \quad 0 \quad 0$$

$$\text{Final conc. } (1 - 0.4) \text{ m} \quad 4 \times 0.4 \quad 0.4 \text{ m} \\ = 0.6 \text{ m} \quad = 1.6 \text{ m}$$

$$\text{Effective molality} = 0.6 + 1.6 + 0.4 = 2.6 \text{ m}$$

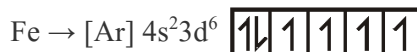
For same boiling point, the molality of another solution should also be 2.6 m.

Now, 18.1 weight percent solution means 18.1 gm solute is present in 100 gm solution and hence, $(100 - 18.1) = 81.9$ gm water.

$$\text{Now, } 2.6 = \frac{18.1/\text{M}}{81.9/1000}$$

$$\therefore \text{Molar mass of solute, } M = 85$$

6. **Ans (49)**



Number of unpaired $e^- = 4$

$$\mu = \sqrt{4(4+2)} \text{ B.M.}$$

$$\mu = \sqrt{24} \text{ B.M.}$$

$$\mu = 4.89 \text{ B.M.}$$

$$\mu = 48.9 \times 10^{-1} \text{ B.M.}$$

Nearest integer value will be 49.

7. **Ans (1)**

$$PV = nRT$$

$$1.0 \times \frac{20}{1000} = \frac{N}{6.023 \times 10^{23}} \times 0.083 \times 273$$

$$\therefore \text{Number of } \text{Cl}_2 \text{ molecules, } N = 5.3 \times 10^{20}$$

$$\text{Hence, Number of Cl-atoms} = 1.06 \times 10^{21} \approx 1 \times 10^{21}$$

8. **Ans (5)**

$$1 \text{ mole KBr (} = 119 \text{ gm)} \text{ have } \frac{10^{-5}}{100} \text{ moles}$$

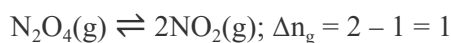
SrBr_2 and hence, 10^{-7} moles cation vacancy

(as 1 Sr^{2+} will result 1 cation vacancy)

$\therefore$ Required number of cation vacancies

$$= \frac{10^{-7} \times 6.023 \times 10^{23}}{119} = 5.06 \times 10^{14} \approx 5 \times 10^{14}$$

9. **Ans (354)**

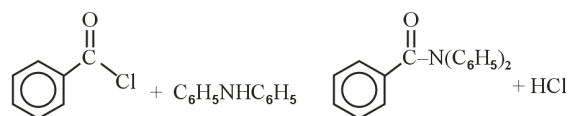


$$\text{Now, } K_p = K_c \cdot (RT)^{\Delta n_g}$$

$$\text{or, } 600.1 = 20.4 \times (0.0831 \times T)^1$$

$$\therefore T = 353.99 \text{ K} = 354 \text{ K}$$

10. Ans (77)



1 mole 1 mole 1 mole
= 140.5 gm = 169 gm = 273 gm

$$\therefore 0.140 \text{ gm} \quad \frac{169}{140.5} \times 0.140$$

$$\text{L.R.} = 0.168 \text{ gm} < 0.388 \text{ gm}$$

excess

$\therefore$ Theoretical amount of given product formed

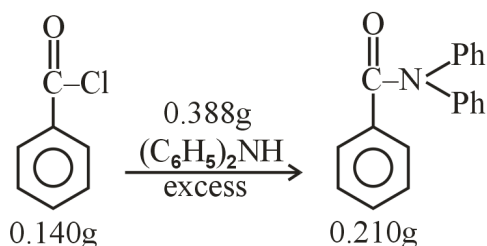
$$= \frac{273}{140.5} \times 0.140 = 0.272 \text{ gm}$$

But its actual amount formed is 0.210 gm.

Hence, the percentage yield of product.

$$= \frac{0.210}{0.272} \times 100 = 77.20 \approx 77$$

OR



$$\text{Mole of Ph-COCl} = \frac{0.140}{140} = 10^{-3} \text{ mol}$$

Mole of $\text{Ph-CO-N}(\text{Ph})_2$, that should be obtained by

mol-mol analysis = 10^{-3} mol.

$$\text{Theoretical mass of product} = 10^{-3} \times 273 = 273 \times 10^{-3} \text{ g}$$

$$\text{Observed mass of product} = 210 \times 10^{-3} \text{ g}$$

$$\% \text{ yield of product} = \frac{210 \times 10^{-3}}{273 \times 10^{-3}} \times 100$$

$$= 76.9\% \approx 77$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

$$f(x) = e^{-x} \sin x$$

$$\text{Now, } F(x) = \int_0^x f(t) dt \Rightarrow F'(x) = f(x)$$

$$I = \int_0^1 (F'(x) + f(x)) e^x dx = \int_0^1 (f(x) + f(x)) \cdot e^x dx$$

$$= 2 \int_0^1 f(x) \cdot e^x dx = 2 \int_0^1 e^{-x} \sin x \cdot e^x dx$$

$$= 2 \int_0^1 \sin x dx$$

$$= 2(1 - \cos 1)$$

$$I = 2 \left\{ 1 - \left(1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{6} + \frac{1}{8} \dots \right) \right\}$$

$$I = 1 - \frac{2}{4} + \frac{2}{6} - \frac{2}{9} + \dots$$

$$1 - \frac{2}{4} < I < 1 - \frac{2}{4} + \frac{2}{6}$$

$$\frac{11}{12} < I < \frac{331}{360}$$

$$\Rightarrow I \in \left[\frac{11}{12}, \frac{331}{360} \right]$$

$$\Rightarrow I \in \left[\frac{330}{360}, \frac{331}{360} \right]$$

2. Ans (A)

$$\text{Let } I = \int_0^{10} \frac{[\sin 2\pi x]}{e^x - [x]} dx = \int_0^{10} \frac{[\sin 2\pi x]}{e^{\{x\}}} dx$$

Function $f(x) = \frac{[\sin 2\pi x]}{e^{\{x\}}}$ is periodic with period '1'

Therefore

$$\begin{aligned} I &= 10 \int_0^1 \frac{[\sin 2\pi x]}{e^{\{x\}}} dx \\ &= 10 \int_0^1 \frac{[\sin 2\pi x]}{e^x} dx \\ &= 10 \left(\int_0^{1/2} \frac{[\sin 2\pi x]}{e^x} dx + \int_{1/2}^1 \frac{[\sin 2\pi x]}{e^x} dx \right) \\ &= 10 \left(0 + \int_{1/2}^1 \frac{(-1)}{e^x} dx \right) \\ &= -10 \int_{1/2}^1 e^{-x} dx \\ &= 10 (e^{-1} - e^{-1/2}) \end{aligned}$$

Now,

$$10 \cdot e^{-1} - 10 \cdot e^{-1/2} = \alpha e^{-1} + \beta e^{-1/2} + \gamma \quad (\text{given})$$

$$\Rightarrow \alpha = 10, \beta = -10, \gamma = 0$$

$$\Rightarrow \alpha + \beta + \gamma = 0$$

3. Ans (B)

$$\cos x (3 \sin x + \cos x + 3) dy$$

$$= (1 + y \sin x (3 \sin x + \cos x + 3)) dx$$

$$\frac{dy}{dx} - (\tan x)y = \frac{1}{(3 \sin x + \cos x + 3) \cos x}$$

$$I.F. = e^{\int -\tan x dx} = e^{\ln|\cos x|} = |\cos x|$$

$$= \cos x \quad \forall x \in \left[0, \frac{\pi}{2}\right)$$

Solution of D.E.

$$y(\cos x) = \int (\cos x) \cdot \frac{1}{\cos x (3 \sin x + \cos x + 3)} dx + C$$

$$y(\cos x) = \int \frac{dx}{3 \sin x + \cos x + 3} dx + C$$

$$y(\cos x) = \int \frac{(\sec^2 \frac{x}{2})}{2 \tan^2 \frac{x}{2} + 6 \tan \frac{x}{2} + 4} dx + C$$

Now

$$\text{Let } I_1 = \int \frac{(\sec^2 \frac{x}{2})}{2 (\tan^2 \frac{x}{2} + 3 \tan \frac{x}{2} + 2)} dx + C$$

$$\text{Put } \tan \frac{x}{2} = t \Rightarrow \frac{1}{2} \sec^2 \frac{x}{2} dx = dt$$

$$I_1 = \int \frac{dt}{t^2 + 3t + 2} = \int \frac{dt}{(t+2)(t+1)}$$

$$= \int \left(\frac{1}{t+1} - \frac{1}{t+2} \right) dt$$

$$= \ln \left| \left(\frac{t+1}{t+2} \right) \right| = \ln \left| \left(\frac{\tan \frac{x}{2} + 1}{\tan \frac{x}{2} + 2} \right) \right|$$

So solution of D.E.

$$y(\cos x) = \ln \left| \frac{1 + \tan \frac{x}{2}}{2 + \tan \frac{x}{2}} \right| + C$$

$$\Rightarrow y(\cos x) = \ln \left(\frac{1 + \tan \frac{x}{2}}{2 + \tan \frac{x}{2}} \right) + C \quad \text{for } 0 \leq x < \frac{\pi}{2}$$

Now, it is given $y(0) = 0$

$$\Rightarrow 0 = \ln \left(\frac{1}{2} \right) + C \Rightarrow \boxed{C = \ln 2}$$

$$\Rightarrow y(\cos x) = \ln \left(\frac{1 + \tan \frac{x}{2}}{2 + \tan \frac{x}{2}} \right) + \ln 2$$

$$\text{For } x = \frac{\pi}{3}$$

$$y\left(\frac{1}{2}\right) = \ln \left(\frac{1 + \frac{1}{\sqrt{3}}}{2 + \frac{1}{\sqrt{3}}} \right) + \ln 2$$

$$y = 2 \ln \left(\frac{2\sqrt{3} + 10}{11} \right)$$

4. **Ans (D)**

$$\sum_{r=0}^6 {}^6C_r \cdot {}^6C_{6-r}$$

$$= {}^6C_0 \cdot {}^6C_6 + {}^6C_1 \cdot {}^6C_5 + \dots + {}^6C_6 \cdot {}^6C_0$$

Now,

$$(1+x)^6(1+x)^6$$

$$= ({}^6C_0 + {}^6C_1x + {}^6C_2x^2 + \dots + {}^6C_6x^6)$$

$$({}^6C_0 + {}^6C_1x + {}^6C_2x^2 + \dots + {}^6C_6x^6)$$

Comparing coefficient of x^6 both sides

$${}^6C_0 \cdot {}^6C_6 + {}^6C_1 + {}^6C_5 + \dots + {}^6C_6 \cdot {}^6C_0 = {}^{12}C_6$$

$$= 924$$

5. **Ans (A)**

We know that

$$r \leq [r] < r + 1$$

$$\text{and } 2r \leq [2r] < 2r + 1$$

$$3r \leq [3r] < 3r + 1$$

$$\vdots \quad \vdots \quad \vdots$$

$$nr \leq [nr] < nr + 1$$

$$r + 2r + \dots + nr$$

$$\leq [r] + [2r] + \dots + [nr] < (r + 2r + \dots + nr) + n$$

$$\frac{\frac{n(n+1)}{2} \cdot r}{n^2} \leq \frac{[r] + [2r] + \dots + [nr]}{n^2} < \frac{\frac{n(n+1)}{2}r + n}{n^2}$$

Now,

$$\lim_{n \rightarrow \infty} \frac{n(n+1) \cdot r}{2 \cdot n^2} = \frac{r}{2}$$

$$\text{and } \lim_{n \rightarrow \infty} \frac{\frac{n(n+1)r}{2} + n}{n^2} = \frac{r}{2}$$

So, by Sandwich Theorem, we can conclude that

$$\lim_{n \rightarrow \infty} \frac{[r] + [2r] + \dots + [nr]}{n^2} = \frac{r}{2}$$

6. **Ans (B)**

Given equation

$$\sin^{-1} \left[x^2 + \frac{1}{3} \right] + \cos^{-1} \left[x^2 - \frac{2}{3} \right] = x^2$$

Now, $\sin^{-1} \left[x^2 + \frac{1}{3} \right]$ is defined if

$$-1 \leq x^2 + \frac{1}{3} < 2 \Rightarrow \frac{-4}{3} \leq x^2 < \frac{5}{3}$$

$$\Rightarrow 0 \leq x^2 < \frac{5}{3} \quad \dots(1)$$

and $\cos^{-1} \left[x^2 - \frac{2}{3} \right]$ is defined if

$$-1 \leq x^2 - \frac{2}{3} < 2 \Rightarrow \frac{-1}{3} \leq x^2 < \frac{8}{3}$$

$$\Rightarrow 0 \leq x^2 < \frac{8}{3} \quad \dots(2)$$

So, from (1) and (2) we can conclude

$$0 \leq x^2 < \frac{5}{3}$$

Case - I if $0 \leq x^2 < \frac{2}{3}$

$$\sin^{-1}(0) + \cos^{-1}(-1) = x^2$$

$$\Rightarrow x + \pi = x^2$$

$$\Rightarrow x^2 = \pi$$

$$\text{but } \pi \notin \left[0, \frac{2}{3} \right)$$

$\Rightarrow$ No value of 'x'

Case - II if $\frac{2}{3} \leq x^2 < \frac{5}{3}$

$$\sin^{-1}(1) + \cos^{-1}(0) = x^2$$

$$\Rightarrow \frac{\pi}{2} + \frac{\pi}{2} = x^2$$

$$\Rightarrow x^2 = \pi$$

$$\text{but } \pi \notin \left[\frac{2}{3}, \frac{5}{3} \right)$$

$\Rightarrow$ No value of 'x'

So, number of solutions of the equation is zero.

7. **Ans (D)**

$$\begin{matrix} 1 & 0 & 0 & 1 \\ \text{odd place} & \text{even place} & \text{odd place} & \text{even place} \end{matrix}$$

$$\text{or } \begin{matrix} 1 & 0 & 0 & 1 \\ \text{even place} & \text{odd place} & \text{even place} & \text{odd place} \end{matrix}$$

$$\Rightarrow \left(\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{3} \right) + \left(\frac{2}{2} \cdot \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \right)$$

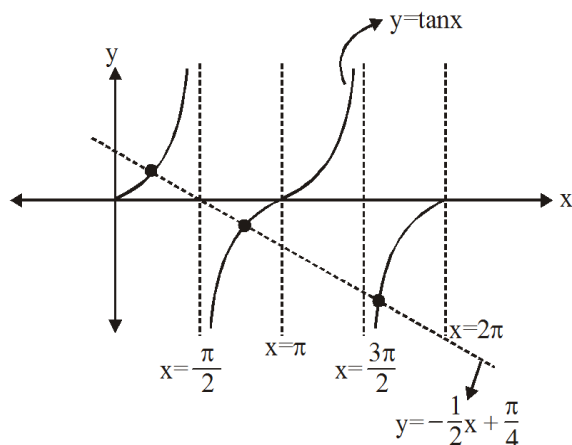
$$\Rightarrow \frac{1}{9}$$

8. Ans (A)

$$x + 2 \tan x = \frac{\pi}{2}$$

$$\Rightarrow 2 \tan x = \frac{\pi}{2} - x$$

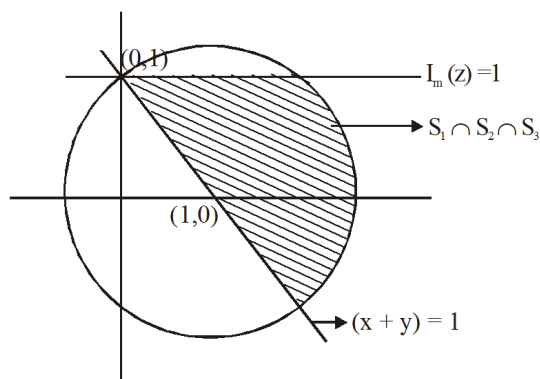
$$\Rightarrow \tan x = -\frac{1}{2}x + \frac{\pi}{4}$$



Number of solutions of the given equation is '3'.

9. Ans (C)

For $|z - 1| \leq \sqrt{2}$, z lies on and inside the circle of radius $\sqrt{2}$ units and centre $(1, 0)$.



For S_2

Let $z = x + iy$

Now, $(1 - i)(z) = (1 - i)(x + iy)$

$\text{Re}((1 - i)z) = x + y$

$\Rightarrow x + y \geq 1$

$\Rightarrow S_1 \cap S_2 \cap S_3$ has infinity many elements

10. Ans (C)

$$\frac{dy}{dx} - \frac{y}{2x} = \frac{x^{9/4}}{x^{5/4}(x^{3/4} + 1)}$$

$$\text{IF} = e^{-\int \frac{dx}{2x}} = e^{-\frac{1}{2} \ln x} = \frac{1}{x^{1/2}}$$

$$y \cdot x^{-1/2} = \int \frac{x^{9/4} \cdot x^{-1/2}}{x^{5/4}(x^{3/4} + 1)} dx$$

$$\int \frac{x^{1/2}}{(x^{3/4} + 1)} dx$$

$$x = t^4 \Rightarrow dx = 4t^3 dt$$

$$\int \frac{t^2 \cdot 4t^3 dt}{(t^3 + 1)}$$

$$4 \int \frac{t^2 (t^3 + 1 - 1)}{(t^3 + 1)} dt$$

$$4 \int t^2 dt - 4 \int \frac{t^2}{t^3 + 1} dt$$

$$\frac{4t^3}{3} - \frac{4}{3} \ln(t^3 + 1) + C$$

$$y x^{-1/2} = \frac{4x^{3/4}}{3} - \frac{4}{3} \ln(x^{3/4} + 1) + C$$

$$1 - \frac{4}{3} \log_e 2 = \frac{4}{3} - \frac{4}{3} \log_e 2 + C$$

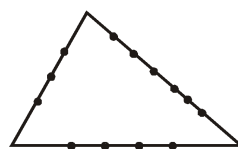
$$\Rightarrow C = -\frac{1}{3}$$

$$y = \frac{4}{3} x^{5/4} - \frac{4}{3} \sqrt{x} \ln(x^{3/4} + 1) - \frac{\sqrt{x}}{3}$$

$$y(16) = \frac{4}{3} \times 32 - \frac{4}{3} \times 4 \ln 9 - \frac{4}{3}$$

$$= \frac{124}{3} - \frac{32}{3} \ln 3 = 4 \left(\frac{31}{3} - \frac{8}{3} \ln 3 \right)$$

11. Ans (C)



Total Number of triangles formed

$$= {}^{14}C_3 - {}^3C_3 - {}^5C_3 - {}^6C_3$$

$$= 333$$

12. Ans (A)

$$\begin{vmatrix} 3 & 4\sqrt{2} & x \\ 4 & 5\sqrt{2} & y \\ 5 & k & z \end{vmatrix} = 0$$

$$R_2 \rightarrow R_1 + R_3 - 2R_2$$

$$\Rightarrow \begin{vmatrix} 3 & 4\sqrt{2} & x \\ 0 & k - 6\sqrt{2} & 0 \\ 5 & k & z \end{vmatrix} = 0$$

$$\Rightarrow (k - 6\sqrt{2})(3z - 5x) = 0$$

$$\text{if } 3z - 5x = 0 \Rightarrow 3(x + 2d) - 5x = 0$$

$$\Rightarrow x = 3d \text{ (Not possible)}$$

$$\Rightarrow k = 6\sqrt{2} \Rightarrow k^2 = 72$$

13. Ans (B)

$$\vec{OP} \perp \vec{OQ}$$

$$\Rightarrow -x + 2y - 3x = 0$$

$$\Rightarrow y = 2x \quad \dots (i)$$

$$|\vec{PQ}|^2 = 20$$

$$\Rightarrow (x+1)^2 + (y-2)^2 + (1+3x)^2 = 20$$

$$\Rightarrow x = 1$$

$$\vec{OP}, \vec{OQ}, \vec{OR} \text{ are coplanar.}$$

$$\Rightarrow \begin{vmatrix} x & y & -1 \\ -1 & 2 & 3x \\ 3 & z & -7 \end{vmatrix} = 0$$

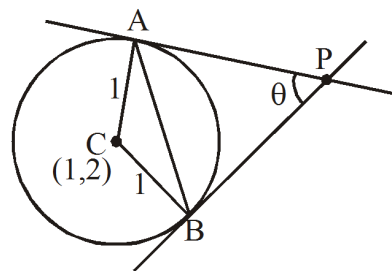
$$\Rightarrow \begin{vmatrix} 1 & 2 & -1 \\ -1 & 2 & 3 \\ 3 & z & -7 \end{vmatrix} = 0$$

$$\Rightarrow 1(-14 - 3z) - 2(7 - 9) - 1(-z - 6) = 0$$

$$\Rightarrow z = -2$$

$$\therefore x^2 + y^2 + z^2 = 1 + 4 + 4 = 9$$

14. Ans (B)



$$\tan \theta = \frac{12}{5}$$

$$PA = \cot \frac{\theta}{2}$$

$$\therefore \text{area of } \triangle PAB = \frac{1}{2} (PA)^2 \sin \theta = \frac{1}{2} \cot^2 \frac{\theta}{2} \sin \theta$$

$$= \frac{1}{2} \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right) \sin \theta$$

$$= \frac{1}{2} \left(\frac{1 + \frac{5}{13}}{1 - \frac{5}{13}} \right) \left(\frac{12}{13} \right) = \frac{1}{2} \frac{18}{18} \times \frac{2}{13} = \frac{27}{26}$$

$$\text{area of } \triangle CAB = \frac{1}{2} \sin \theta = \frac{1}{2} \left(\frac{12}{13} \right) = \frac{6}{13}$$

$$\therefore \frac{\text{area of } \triangle PAB}{\text{area of } \triangle CAB} = \frac{9}{4}$$

15. Ans (B)

$$f(x) = \begin{cases} -x \left(2 - \sin \left(\frac{1}{x} \right) \right) & x < 0 \\ 0 & x = 0 \\ x \left(2 - \sin \left(\frac{1}{x} \right) \right) & x > 0 \end{cases}$$

$$f'(x) = \begin{cases} - \left(2 - \sin \frac{1}{x} \right) - x \left(-\cos \frac{1}{x} \cdot \left(-\frac{1}{x^2} \right) \right) & x < 0 \\ \left(2 - \sin \frac{1}{x} \right) + x \left(-\cos \frac{1}{x} \cdot \left(-\frac{1}{x^2} \right) \right) & x > 0 \end{cases}$$

$$f'(x) = \begin{cases} -2 + \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x} & x < 0 \\ 2 - \sin \frac{1}{x} + \frac{1}{x} \cos \frac{1}{x} & x > 0 \end{cases}$$

$f'(x)$ is an oscillating function which is

non-monotonic in $(-\infty, 0) \cup (0, \infty)$.

16. Ans (B)

Tangent to parabola

$$2y = 2(x+6) - 20$$

$$\Rightarrow y = x - 4$$

Condition of tangency for ellipse.

$$16 = 2(1)^2 + b$$

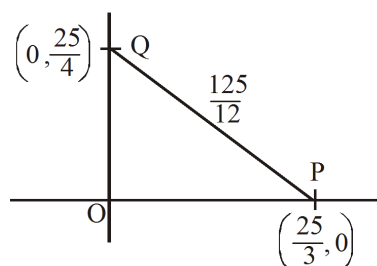
$$\Rightarrow b = 14$$

17. Ans (A)

$$\begin{aligned} & \lim_{\theta \rightarrow 0} \frac{\tan(\pi(1 - \sin^2 \theta))}{\sin(2\pi \sin^2 \theta)} \\ &= \lim_{\theta \rightarrow 0} \frac{-\tan(\pi \sin^2 \theta)}{\sin(2\pi \sin^2 \theta)} \\ &= \lim_{\theta \rightarrow 0} - \left(\frac{\tan(\pi \sin^2 \theta)}{\pi \sin^2 \theta} \right) \left(\frac{2\pi \sin^2 \theta}{\sin(2\pi \sin^2 \theta)} \right) \times \frac{1}{2} \\ &= \frac{-1}{2} \end{aligned}$$

18. Ans (C)

Tangent to circle $3x + 4y = 25$



$$OP + OQ + OR = 25$$

$$\begin{aligned} \text{Incentre} &= \left(\frac{\frac{25}{4} \times \frac{25}{3}}{25}, \frac{\frac{25}{4} \times \frac{25}{3}}{25} \right) \\ &= \left(\frac{25}{12}, \frac{25}{12} \right) \\ \therefore r^2 &= 2 \left(\frac{25}{12} \right)^2 = 2 \times \frac{625}{144} = \frac{625}{72} \end{aligned}$$

19. Ans (A)

Option (1)

$$(p \wedge q) \rightarrow (p \rightarrow q)$$

$$= \sim (p \wedge q) \vee (\sim p \vee q)$$

$$= (\sim p \vee \sim q) \vee (\sim p \vee q)$$

$$= \sim p \vee (\sim q \vee q)$$

$$= \sim p \vee t$$

$$= t$$

Option (2)

$$(p \wedge q) \wedge (p \vee q) = (p \wedge q) \quad (\text{Not a tautology})$$

Option (3)

$$(p \wedge q) \vee (p \rightarrow q)$$

$$= (p \wedge q) \vee (\sim p \vee q)$$

$$= \sim p \vee q \quad (\text{Not a tautology})$$

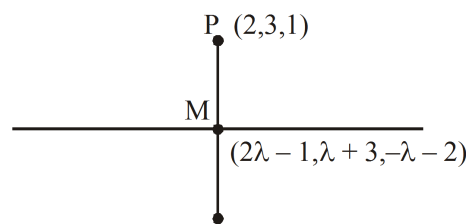
Option (4)

$$= (p \wedge q) \wedge (\sim p \vee q)$$

$$= p \wedge q \quad (\text{Not a tautology})$$

20. Ans (B)

$$\text{Line } \frac{x+1}{2} = \frac{y-3}{1} = \frac{z+2}{-1}$$



$$\overrightarrow{PM} = (2\lambda - 3, \lambda, -\lambda - 3)$$

$$\overrightarrow{PM} \perp (2\hat{i} + \hat{j} - \hat{k})$$

$$4\lambda - 6 + \lambda + \lambda + 3 = 0 \Rightarrow \lambda = \frac{1}{2}$$

$$\therefore M \equiv \left(0, \frac{7}{2}, \frac{-5}{2} \right)$$

$\therefore$ Reflection $(-2, 4, -6)$

$$\text{Plane : } \begin{vmatrix} x-2 & y-1 & z+1 \\ 3 & -2 & 1 \\ 4 & -3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(-10+3) - (y-1)(15-4) + (z+1)(-1) = 0$$

$$\Rightarrow -7x + 14 - 11y + 11 - z - 1 = 0$$

$$\Rightarrow 7x + 11y + z = 24$$

$$\therefore \alpha = 7, \beta = 11, \gamma = 1$$

$$\alpha + \beta + \gamma = 19$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2)

$$2\log_{10}(4^x - 2) = 1 + \log_{10}\left(4^x + \frac{18}{5}\right)$$

$$(4^x - 2)^2 = 10\left(4^x + \frac{18}{5}\right)$$

$$(4x)^2 + 4 - 4(4^x) - 32 = 0$$

$$(4^x - 16)(4^x + 2) = 0$$

$$4^x = 16$$

$$x = 2$$

$$\begin{vmatrix} 3 & 1 & 4 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix} = 3(-2) - 1(0 - 4) + 4(1)$$

$$= -6 + 4 + 4 = 2$$

2. Ans (5)

$$f: [-1, 1] \rightarrow \mathbb{R}$$

$$f(x) = ax^2 + bx + c$$

$$f(-1) = a - b + c = 2 \quad \dots(1)$$

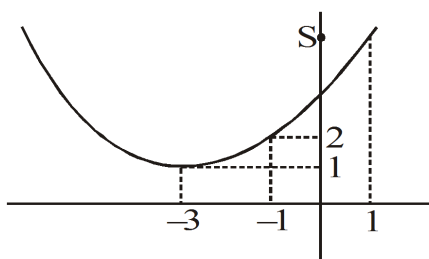
$$f'(-1) = -2a + b = 1 \quad \dots(2)$$

$$f''(x) = 2a$$

$$\Rightarrow \text{Max. value of } f''(x) = 2a = \frac{1}{2}$$

$$\Rightarrow a = \frac{1}{4}; b = \frac{3}{2}; c = \frac{13}{4}$$

$$\therefore f(x) = \frac{x^2}{4} + \frac{3}{2}x + \frac{13}{4}$$



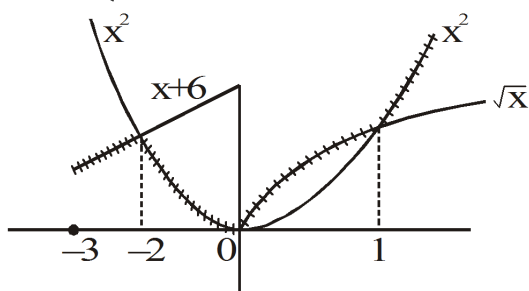
$$\text{For, } x \in [-1, 1] \Rightarrow 2 \leq f(x) \leq 5$$

$$\therefore \text{Least value of } a \text{ is } 5$$

3. Ans (41)

$$f: [-3, 1] \rightarrow \mathbb{R}$$

$$f(x) = \begin{cases} \min \{ (x+6), x^2 \} & , -3 \leq x \leq 0 \\ \max \{ \sqrt{x}, x^2 \} & , 0 \leq x \leq 1 \end{cases}$$



area bounded by $y = f(x)$ and x -axis

$$= \int_{-3}^{-2} (x+6) dx + \int_{-2}^0 x^2 dx + \int_0^1 \sqrt{x} dx$$

$$A = \frac{41}{6}$$

$$6A = 41$$

4. Ans (144)

Since orthocentre and circumcentre both lie on y -axis

$\Rightarrow$ Centroid also lies on y -axis

$$\Rightarrow \sum \cos \alpha = 0$$

$$\cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\Rightarrow \cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma = 3 \cos \alpha \cos \beta \cos \gamma$$

$$\therefore \frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos \alpha \cos \beta \cos \gamma}$$

$$= \frac{4(\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma) - 3(\cos \alpha + \cos \beta + \cos \gamma)}{\cos \alpha \cos \beta \cos \gamma}$$

$$= 12$$

5. Ans (68)

Let number be $a_1, a_2, a_3, \dots, a_{2n}, b_1, b_2, b_3, \dots, b_n$

$$\sigma^2 = \frac{\sum a^2 + \sum b^2}{3n} - (5)^2$$

$$\Rightarrow \sum a^2 + \sum b^2 = 87n$$

Now, distribution becomes

$$a_1 + 1, a_2 + 1, a_3 + 1, \dots, a_{2n} + 1, b_1 - 1,$$

$$b_2 - 1, \dots, b_n - 1$$

Variance

$$= \frac{\sum (a+1)^2 + \sum (b-1)^2}{3n} - \left(\frac{12n+2n+3n-n}{3n} \right)^2$$

$$= \frac{(\sum a^2 + 2n + 2 \sum a) + (\sum b^2 + n - 2 \sum b)}{3n}$$

$$= \frac{(\sum a^2 + 2n + 2 \sum a) + (\sum b^2 + n - 2 \sum b)}{3n} - \left(\frac{16}{3} \right)^2$$

$$= \frac{87n + 3n + 2(12n) - 2(3n)}{3n} - \left(\frac{16}{3} \right)^2$$

$$\Rightarrow k = \frac{108}{3} - \left(\frac{16}{5} \right)^2$$

$$\Rightarrow 9k = 3(108) - (16)^2 = 324 - 256 = 68$$

6. Ans (4)

$$T_{r+1} = {}^nC_r (x)^{n-r} \left(\frac{a}{x^2} \right)^r$$

$$= {}^nC_r a^r x^{n-3r}$$

$${}^nC_2 a^2 : {}^nC_3 a^3 : {}^nC_4 a^4 = 12 : 8 : 3$$

After solving

$$n = 6, a = \frac{1}{2}$$

$$\text{For term independent of 'x' } \Rightarrow n = 3r$$

$$r = 2$$

$$\therefore \text{Coefficient is } {}^6C_2 \left(\frac{1}{2} \right)^2 = \frac{15}{4}$$

Nearest integer is 4.

7. **Ans (2020)**

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$AB = B$$

$$\Rightarrow (A - I) B = O$$

$$\Rightarrow |A - I| = 0, \text{ since } B \neq O$$

$$\begin{vmatrix} (a-1) & b \\ c & (d-1) \end{vmatrix} = 0$$

$$ad - bc = 2020$$

8. **Ans (486)**

Let $\vec{x} = \lambda \vec{a} + \mu \vec{b}$ (λ and μ are scalars)

$$\vec{x} = \hat{i} (2\lambda + \mu) + \hat{j} (2\mu - \lambda) + \hat{k} (\lambda - \mu)$$

$$\text{Since } \vec{x} \cdot (3\hat{i} + 2\hat{j} - \hat{k}) = 0$$

$$3\lambda + 8\mu = 0 \quad \dots(1)$$

Also Projection of $\vec{x}$ on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$

$$\frac{\vec{x} \cdot \vec{a}}{|\vec{a}|} = \frac{17\sqrt{6}}{2}$$

$$6\lambda - \mu = 51 \quad \dots(2)$$

From (1) and (2)

$$\lambda = 8, \mu = -3$$

$$\vec{x} = 13\hat{i} - 14\hat{j} + 11\hat{k}$$

$$|\vec{x}|^2 = 486$$

9. **Ans (1)**

$$I_n = \int_1^e x^{19} (\log|x|)^n dx$$

$$I_n = \left| (\log|x|)^{19} \frac{x^{20}}{20} \right|_1^e - \int_1^e n (\log|x|)^{n-1} \cdot \frac{1}{x} \cdot \frac{x^{20}}{20} dx$$

$$20I_n = e^{20} - nI_{n-1}$$

$$\therefore 20I_{10} = e^{20} - 10I_9$$

$$20I_9 = e^{20} - 9I_8$$

$$\Rightarrow 20I_{10} = 10I_9 + 9I_8$$

$$\alpha = 10, \beta = 9$$

10. **Ans (0)**

Let point P is (α, β, γ)

$$\left(\frac{\alpha + \beta + \gamma}{\sqrt{3}} \right)^2 + \left(\frac{\ell\alpha - n\gamma}{\sqrt{\ell^2 + n^2}} \right)^2 + \left(\frac{\alpha - 2\beta + \gamma}{\sqrt{6}} \right)^2 = 9$$

Locus is

$$\frac{(x + y + z)^2}{3} + \frac{(\ell x - nz)^2}{\ell^2 + n^2} + \frac{(x - 2y + z)^2}{6} = 9$$

$$x^2 \left(\frac{1}{2} + \frac{\ell^2}{\ell^2 + n^2} \right) + y^2 + z^2 \left(\frac{1}{2} + \frac{n^2}{\ell^2 + n^2} \right) +$$

$$2zx \left(\frac{1}{2} - \frac{\ell n}{\ell^2 + n^2} \right) - 9 = 0$$

Since its given that $x^2 + y^2 + z^2 = 9$

After solving $\ell = n$